BISHOWDEEP BHUSAL

Self-Taught Builder | Hardware | Python | IoT

ABOUT

16-year-old self-taught builder. Started with a broken Arduino at 13. Built my first robot at 14 with a budget of ■3,000 and zero formal training. I learn by doing, break things to understand them, and never stop asking why. No mentors, no classes — just curiosity, YouTube, and an obsession with how things work.

5	4	8+	■3K
Robots Built	ESP32s Destroyed	Months Coding	Robot Budget

PROJECTS

ESP32 AI Robot That Talks

AI + Hardware

Built a robot that can see, hear, and speak using ESP32 + Gemini Flash API. Camera sensor captures frames, mic module listens for voice commands, and the AI processes everything in real-time. Motor driver controls movement. The robot answers questions, recognizes objects, and responds to voice.

ESP32, Gemini Flash API, Camera Module, Mic Module, Motor Driver, WiFi

AI-powered | Voice + Vision | Real-time

ESP32 Web-Controlled Robot

Hardware

Built a robot that hosts its own Wi-Fi network and serves a control interface from any browser. Motor driver integration, web server on ESP32, responsive controls. The project that started everything.

ESP32, Arduino, Web Server, Wi-Fi AP, Motor Driver

■3K budget | 2 weeks build | 4 broken ESP32s

Course Recommendation Engine

Software

Matrix-based recommendation system in Python. 24 courses across 4 categories. Takes grade, interest, time, seriousness — outputs personalized picks. Discovered 2D arrays independently while learning.

Python, 2D Arrays, Matrices, Input Logic

5 iterations | 2 hours total | 24 courses

Smart Home Automation

IoT

Arduino-based home automation with relay modules. Control lights and fans from a phone via custom web interface. Learned about relay switching, AC wiring basics, and web-to-hardware communication.

Arduino Mega, Relay Module, Web Interface, WiFi

4 channels | 1 week build | Phone control

Self-Learning Experiments

Research

Tracked my own learning patterns. Juggling: 0→5→10→13 catches over 3 days. Proved step-function learning — struggle, plateau, regression, breakthrough. Nap was critical for consolidation.

Self-Experiment, Motor Learning, Data Tracking

3 days | 13 max catches | 0→13 growth

SKILLS

Hardware & Electronics

ESP32, Arduino, motor drivers, soldering, multimeter, circuit debugging, sensor integration

Python

Variables, loops, conditionals, 2D arrays, matrices, input handling, API basics

C++ / Arduino

GPIO control, motor protocols, serial communication, sensor libraries, basic embedded

Web Basics

HTML/CSS, ESP32 web servers, browser-based control, responsive interfaces

AI & APIs

API integration concepts, AI embedding research, using ChatGPT/Claude for coding help

Problem Solving

Debugging, first-principles thinking, rapid prototyping, learning from failure, iteration

CERTIFICATIONS & LEARNING

Certificate	Provider	Status
CS50: Intro to Computer Science	Harvard — edX	**Completed**
Python for Everybody	U of Michigan — Coursera	**In Progress**
Arduino Programming Fundamentals	Microsoft Learn	**In Progress**
Intro to Electronics	Khan Academy	**Queued**
Scientific Computing with Python	freeCodeCamp	**Queued**
Raspberry Pi Physical Computing	Raspberry Pi Foundation	**Queued**

CONTACT

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Always looking for the next challenge. If it involves building, learning, or breaking things to understand them — I'm in.